

Midterm 1 (Ch1-7, HW1-5) 7-8:30pm Sayles Auditorium -- individual score

● Graded

Student

Baidy Mbaye

Total Points

41 / 75 pts

Question 1

MC1

3 / 3 pts

✓ - 0 pts Correct

Question 2

MC2

0 / 3 pts

✓ - 3 pts Incorrect, answer is C

Question 3

MC3

3 / 3 pts

✓ - 0 pts Correct

Question 4

MC4

3 / 3 pts

✓ - 0 pts Correct

Question 5

MC5

3 / 3 pts

✓ - 0 pts Correct

Question 6

MC6

0 / 3 pts

✓ - 3 pts Incorrect, answer is D

Question 7

MC7

3 / 3 pts

✓ - 0 pts Correct

Question 8

MC8

0 / 3 pts

✓ - 3 pts Incorrect, answer is D

Question 9

MC9

3 / 3 pts

✓ - 0 pts Correct

Question 10

MC10

0 / 3 pts

✓ - 3 pts Incorrect, answer is B

Question 11

MC11

0 / 3 pts

✓ - 3 pts Incorrect, answer is B

Question 12

MC12

3 / 3 pts

✓ - 0 pts Correct

Question 13

MC13

3 / 3 pts

✓ - 0 pts Correct

Question 14

MC14

0 / 3 pts

✓ - 3 pts Incorrect, answer is A

Question 15

MC15

3 / 3 pts

✓ - 0 pts Correct

Question 16

MC16

3 / 3 pts

✓ - 0 pts Correct

Question 17

MC17

3 / 3 pts

✓ - 0 pts Correct

Question 18

Short Answer 1

3 / 3 pts

✓ - 0 pts Correct answer: $\{\$43.86(1-0.052) [1.011-0.0105-0.0025] - \$43.86\}/\$43.86 = 4.02\%$

Question 19

Short Answer 2

0.5 / 6 pts

✓ - 0 pts Correct answer: lending: $y = 1$ so $(0.10 - 0.035)/A(0.15)^2 = 1$ so $A = 2.89$; borrowing: $y = 1$ so $(0.10 - 0.07)/A(0.15)^2 = 1$ so $A = 1.33$ so Range of A: $1.33 < A < 2.89$

✓ - 6 pts Incorrect

✓ + 0.5 pts Understood needed to do a utility calculation

Question 20

Short Answer 3a

1 / 3 pts

✓ - 0 pts Correct answer: $0.25 = (20800 + 11440 - 100p) / 100p$. A margin call occurs when $p > \$257.92$.

✓ - 2 pts Followed a buying on margin format as opposed to shorting

Question 21

Short Answer 3b

0.5 / 2 pts

✓ - 0 pts Correct. Return = Profit/Cost = $(208 - 150)100 / 11440 = 50.7\%$

✓ - 1.5 pts Got the spirit of the question

Question 22

Short Answer 3c

2 / 2 pts

✓ - 0 pts Correct answer: $(208 \times 100) - (150 \times 100) / (150 \times 100) = 38.7\%$

Question 23

Short Answer 4a

1 / 6 pts

✓ - 0 pts Correct answer: $E(r) = .25(12\%) + .50(5\%) + 0.25(-6\%) = 4.0\%$; variance = $0.25(.12 - .04)^2 + 0.5(0.05 - 0.04)^2 + 0.25(-0.06 - 0.04)^2 = 0.00415$ so std dev = 6.44%; Sharpe ratio = $[E(r_P) - r_f] / \text{std dev} = (0.04 - 0.02) / 0.0633 = 0.31$

✓ - 2 pts Incorrect Expected Return. Expected Return should be 4%

✓ - 2 pts Missing Standard Deviation Formula

✓ - 1 pt Correct Sharpe formula but incorrect inputs

Question 24

Short Answer 4b

0 / 2 pts

✓ - 0 pts Correct answer: $y^* = (0.04 - 0.02) / [4.0(0.0644)^2] = 1.20$ so 120% invested in the risky portfolio and borrow 20% at the risk-free rate.

✓ - 2 pts (essentially) blank

Economics 1710: Investments I
Spring 2026 -- KUO
EXAM #1

NAME: Mbaye, Baidy

GROUP

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Instructions: You have 50 minutes for this exam, so please pace yourself appropriately. Only writing implements and the Econ Department calculator are permitted.

Calculator test: try to solve $\frac{1}{1.04^6}$ and see if you get 0.790314... [many decimal places]

Make sure to INITIAL each page since we will be scanning the exams, in case they get out of order.

In the multiple choice section, make sure to write your answer on the LINE in front of the question number.

In the problem section, to maximize your chances of obtaining partial credit, please show your work neatly as well as the specific formulas used. Note that even if you cannot answer a part, you may be able to answer the following part, or to at least list the formulas you would have used.

Part A.

Multiple choice _____ x3 = _____ (51 pts)
(17 Q = 3 pts each)

Part B. _____/3 = _____ (3 pts)

Short problems _____/6 = _____ (6 pts)

_____/3
_____/2
_____/2 = _____ (7 pts)

_____/6
_____/2 = _____ (8 pts)

total _____ (75 pts total)

Section A: Multiple Choice (3 points each)

E 1) An investor purchased a bond 45 days ago for \$985. He received \$15 in interest and sold the bond for \$980. What is the holding-period return on his investment?

- A. 1.52%
- B. 0.50%
- C. 1.92%
- D. 0.01%
- E. None of the above

$$HPR = \frac{\text{Final} - \text{Initial}}{\text{Initial}} = \frac{985 + 15 - 985}{985} = \frac{10}{985} \approx 0.01\%$$

A 2) The index that includes the largest number of actively traded stock is:

- ☒ A. the NASDAQ Composite Index.
- ☐ B. the NYSE Composite Index.
- ☐ C. the Wilshire 5000 Index.
- ☐ D. the Value Line Composite Index.
- ☐ E. the Russell Index.

B 3) You sell short 100 shares of Loser Co. at a market price of \$45 per share. Your maximum possible loss is

- A. \$4500
- ☒ B. unlimited
- C. zero
- D. \$9000
- E. cannot tell from the information given

D 4) An investment provides a 0.78% return monthly, its effective annual rate is

- A. 9.36%
- B. 9.63%
- C. 10.02%
- ☒ D. 9.77%
- E. none of the above.

$$EAR = \left[1 + \frac{0.78}{12} \right]^{12} - 1$$

$$EAR = \left[1 + \frac{0.78}{12} \right]^{12} - 1$$

$$0.0977 \quad 9.77\%$$

A 5) Historical records regarding return on stocks, Treasury bonds, and Treasury bills between 1926 and 2009 show that

- A. stocks offered investors greater rates of return than bonds and bills.
- B. stock returns were less volatile than those of bonds and bills.
- C. bonds offered investors greater rates of return than stocks and bills.
- D. bills outperformed stocks and bonds.
- E. treasury bills always offered a rate of return greater than inflation.

B 6) When assessing tail risk by looking at the 5% worst-case scenario, the VaR is the _____.

- A. most realistic as it is the most complete measure of risk
- B. most pessimistic as it is the most complete measure of risk
- C. most optimistic as it is the most complete measure of risk
- D. most optimistic as it takes the highest return (smallest loss) of all the cases
- E. none of the above

B 7) According to the mean-variance criterion, which of the statements below is correct?

Investment	E(r)	Standard Deviation
A	10%	5%
B	21%	11%
C	18%	23%
D	24%	16%

$\sigma(5) \cdot \tau(5)$

- A. Investment B dominates Investment A.
- B. Investment B dominates Investment C.
- C. Investment D dominates all of the other investments.
- D. Investment D dominates only Investment B.
- E. Investment C dominates investment A.

A 8) An investor purchases one municipal and one corporate bond that pay rates of return of 6% and 8%, respectively. If the investor is in the 25% marginal tax bracket, his or her after tax rates of return on the municipal and corporate bonds would be _____ and _____, respectively.

- A. 6% and 8%
- B. 4.5% and 6%
- C. 4.5% and 8%
- D. 6% and 6%
- E. None of the above

$$6\% = 8\% - [1 - \text{tax bracket}]$$

$$1 - 25$$

$$.06 = .08 - [1 - .25]$$

$$= .08 - [.75]$$

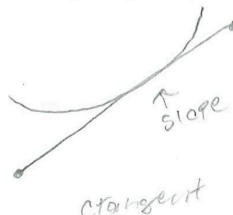
$$8\% + [1 - \text{tax bracket}]$$

$$= .08 - [.75]$$

C 9) The Capital Allocation Line provided by a risk-free security and N risky securities is

- A. the line that connects the risk-free rate and the global minimum-variance portfolio of the risky securities.
- B. the line that connects the risk-free rate and the portfolio of the risky securities that has the highest expected return on the efficient frontier. (slope, most efficient allocation)
- C. the line tangent to the efficient frontier of risky securities drawn from the risk-free rate.
- D. the horizontal line drawn from the risk-free rate.
- E. none of the above.

Sharpe ratio



A 10) You buy 300 shares of Qualitycorp for \$30 per share and deposit initial margin of 50%. The next day Qualitycorp's price drops to \$25 per share. What is your actual margin?

- ☒ A. 50%
- ☐ B. 40%
- ☐ C. 33%
- ☐ D. 60%
- ☐ E. 25%

300x30 initial margin
50% 25 shouldn't change
the amount you loan
will enough

$$\text{Margin} = \frac{\text{total} - \text{Loan}}{\text{total}} \quad 50\% =$$

A 11) When two risky securities that are positively correlated but not perfectly correlated are held in a portfolio,

- ☒ A. the portfolio standard deviation will be greater than the weighted average of the individual security standard deviations.
- ☐ B. the portfolio standard deviation will be less than the weighted average of the individual security standard deviations.
- ☐ C. the portfolio standard deviation will be equal to the weighted average of the individual security standard deviations.
- ☐ D. the portfolio standard deviation will always be equal to the securities' covariance.
- ☐ E. none of the above are true.

σ ρ $0 < \rho < 1$
can't be perfectly hedged = $\rho = 1$ $\rho \sigma_A \times \sigma_B = \text{Cov}(r_1, r_2)$
 $w_A = \frac{\sigma_B}{\sigma_A + \sigma_B}$ $w_B = \frac{\sigma_A}{\sigma_B + \sigma_A}$

B 12) The change from a straight to a kinked capital allocation line is a result of:

- ☐ A. reward-to-volatility ratio increasing.
- ☒ B. borrowing rate exceeding lending rate.
- ☐ C. an investor's risk tolerance decreasing.
- ☐ D. increase in the portfolio proportion of the risk-free asset.
- ☐ E. none of the above.

D 13) Based on their relative degrees of risk tolerance

- ☐ A. investors will hold varying amounts of the risky asset in their portfolios.
- ☐ B. investors will hold varying amounts of the risk-free asset in their portfolios.
- ☐ C. all investors will have the same portfolio asset allocations.
- ☒ D. A and B.
- ☐ E. A and C.

↓
No
investors
will have the
same optimal
portfolios
but not
asset allocations

- C 14) Which of the following statement(s) is (are) false regarding the selection of a portfolio from those that lie on the Capital Allocation Line? $\rightarrow$ most efficient (SLATE)
- A. Less risk-averse investors will invest more in the risk-free security and less in the optimal risky portfolio than more risk-averse investors.
 - B. More risk-averse investors will invest less in the optimal risky portfolio and more in the risk-free security than less risk-averse investors.
 - C. Investors choose the portfolio that maximizes their expected utility.
 - D. A and B.
 - E. A and C.

- E 15) A mutual fund had year-end assets of \$700,000,000 and liabilities of \$7,000,000. There were 40,150,000 shares in the fund at year-end. What was the mutual fund's Net Asset Value?
- $\frac{\text{total} - \text{liabilities}}{\text{outstanding shares}}$
- A. \$9.63
 - B. \$57.71
 - C. \$16.42
 - D. \$17.87
 - E. \$17.26

- C 16) Consider a T-bill with a rate of return of 5 percent and the following risky securities:
- Security A: $E(r) = 0.15$; Variance = 0.04 190
- Security B: $E(r) = 0.10$; Variance = 0.0225 225
- Security C: $E(r) = 0.12$; Variance = 0.01 190
- Security D: $E(r) = 0.13$; Variance = 0.0625 190
- From which set of portfolios, formed with the T-bill and any one of the 4 risky securities, would a risk-averse investor always choose his portfolio?
- A. The set of portfolios formed with the T-bill and security A.
 - B. The set of portfolios formed with the T-bill and security B.
 - C. The set of portfolios formed with the T-bill and security C.
 - D. The set of portfolios formed with the T-bill and security D.
 - E. Cannot be determined.

- B 17) Consider an investment opportunity set formed with two securities that are perfectly negatively correlated. The global minimum variance portfolio has a standard deviation that is always
- $\rho = -1$
- A. greater than zero.
 - B. equal to zero.
 - C. equal to the sum of the securities' standard deviations.
 - D. equal to -1.
 - E. none of the above.
- $\text{variance} = 0 = 0$
if $\rho = -1$
perfectly hedged

initials: BM

Section B: Problems

Question 1: You purchased shares of the Columbia Global Technology Growth Fund (CTCAX) at a price of \$43.86 per share at the beginning of the year and paid a front-end load of 5.20%. If the securities in which the fund invested increased in value by 11.0% during the year, and the fund's expense ratio was 1.05% with a 12b-1 fee of 0.25%, what would your return be if you sold the fund at the end of the year? (3 points)

$43.86 \times 0.948 = 41.58$
 $43.86 \times [1 + 0.0520] = 46.14$
 $46.14 - 43.86 = 2.28$
 $43.86 - 2.28 = 41.58$
 $11\% - [1.05 + .25] = 9.7\%$
 $41.58 \times [1 + 0.097] = 45.61$
 $\frac{45.61 - 43.86}{43.86} = 0.0398$
 $\text{return rate} = 3.98\%$
 $\text{Final Profit} = \$1.75$

that's been compounded
 true annual return rate

Question 2: Suppose you manage a risky portfolio with expected return of 10% and standard deviation of 15%. The risk-free rate is 3.5% and investors may borrow at 7.0%. Suppose investors have the standard mean-variance utility function: $U = E(r) - (A/2)\sigma^2$. What is the range of risk aversion for which investors will neither borrow nor lend, that is, for which $y = 1$? (6 points)

$E(r) = 10\%$ $\sigma = 15\%$ $r_f = 3.5\%$ $r_b = 7\%$

risk free rate = utility
 ↓
 indifference

$y = 1$



$$3.5 = 10 - \frac{1}{2}(A)(15)^2$$

$$.035 = .1 - \frac{1}{2}(A)(.15)^2$$

$$.1 - \frac{1}{2}(.0225)(A)$$

$$.1 - .01125$$

$$.035 = .08875(A)$$

$$A = .394$$

initials: BM

Question 3. (7 points total) Suppose you have a margin account that has a maintenance margin of 25% and a margin interest rate of 10.5%. You decide to sell short 100 shares of Amazon (AMZN) at \$208.00 per share and put down an initial margin of 55%.

(a) For what range of stock prices would you receive a margin call? (3 points)

$$\begin{aligned} \text{initial} &= 55\% \\ 100 \times 208 &= 20,800 \\ .55 &= \frac{20,800 - \text{loan}}{20,800} \\ 11,440 &= 20,800 \\ \text{loan} &= 9,360 \\ 25\% &= \frac{P - 9,360}{P} \\ .25P &= P - 9,360 \\ .25P - P &= -9,360 \\ P(.25 - 1) &= -9,360 \\ P(-.75) &= -9,360 \end{aligned}$$

The price per share would need to fall to \$124.8 or below for a margin call to happen

(b) What would be your rate of return if you repurchase the stock at \$150.00 per share? Amazon pays no dividends, and you did not remove any money from the account before making the offsetting transaction. (2 points)

$$\begin{aligned} 208 - 150 &= 58 \downarrow \\ 150 - 208 &= -58 \\ \text{total profit} &= 58 \\ \text{initial} &= 208 \\ \frac{58}{208} &= .278 \\ &= 27.8\% \end{aligned}$$

35-30 = 5
1500

Sold
bought

(c) Suppose instead you had bought 100 shares of AMZN at \$150 and sold at \$208. What is your rate of return? (2 points)

$$\begin{aligned} \frac{\text{total } P \ 58}{\text{Initial } 150} &= .386 \times 100 \\ &= 38.6\% \\ \text{rate of return} &= 38.6\% \end{aligned}$$

FLIP THE PAGE

initials: BM

Question 4: (8 points total) You have been given this probability distribution for the holding-period return (HPR) for a stock:

<u>State of the Economy</u>	<u>Probability</u>	<u>HPR</u>
Boom	0.25	0.12
Normal growth	0.50	0.05
Recession	0.25	-0.06

(a) Suppose that the risk-free rate is 0.02. What is the Sharpe ratio? (6 points)

$$\frac{E(R) - 0.02}{\sigma} = \frac{0.04}{0.25} = 0.16$$

(b) Suppose you have mean-variance utility so $U = E(r) - \frac{1}{2}A\sigma^2$ and your coefficient of risk aversion were 4.0. What holdings (and in what proportions) comprise your optimal portfolio, using the lending/borrowing terminology? (2 points)

0.4

